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## **Project Report**

# **FDTD METHOD: A COMPLETE TECHNICAL GUIDE TO ELECTROMAGNETIC WAVE SIMULATION**

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## Introduction:

Maxwell's equations describe how electric and magnetic fields behave and interact in space. While these equations are powerful, solving them by hand for real-world problems—like how a radio wave passes through a building or how light scatters off a lens—is extremely difficult. This is where numerical methods come in.

The Finite-Difference Time-Domain (FDTD) method, first developed by Kane Yee in 1966, is one of the most popular ways to solve Maxwell's equations using a computer. Instead of trying to find a single perfect mathematical formula, FDTD breaks space and time into tiny discrete chunks and calculates the fields step-by-step. Think of it like a stop-motion movie: instead of showing continuous motion, we capture one frame at a time. Each frame shows the electric and magnetic fields at every point in space, and by playing thousands of these frames rapidly, we see how electromagnetic waves propagate, reflect, and transmit through different materials.

This report presents a complete technical guide to understanding and implementing the FDTD method. We explain how continuous physics is converted into discrete computer-friendly steps, how the famous Yee Grid arranges electric and magnetic fields in a staggered pattern, and how we analyze the results using Fourier transforms.

## Motivation & Objective:

The primary objective of this project is to develop a deep, intuitive understanding of the FDTD method and its underlying mathematics. While our previous project introduced the basic 1D FDTD simulation, this guide expands the scope to include 2D simulations, proper boundary conditions, stability criteria, and real-time spectral analysis.

By the end of this guide, we aim to:

- Understand how Maxwell's continuous curl equations are converted into simple algebraic update equations.
- Learn how the Yee Grid spatially staggers electric and magnetic fields for accurate computation.
- Implement stable time-stepping using the Courant-Friedrichs-Lewy (CFL) condition.
- Apply Mur Absorbing Boundary Conditions to simulate open, infinite space within a finite computer memory.
- Perform on-the-fly Discrete Fourier Transform (DFT) to extract frequency-domain reflectance and transmittance.

This project bridges the gap between theoretical electromagnetics and practical computational engineering.

## Theory:

Electromagnetic waves are governed by Maxwell's curl equations. In a simple, source-free material, these tell us that a changing electric field creates a magnetic field, and a changing magnetic field creates an electric field. This mutual creation is what allows light and radio waves to travel through space.

### 1. From Continuous to Discrete

In calculus, we use derivatives to describe rates of change. But computers cannot handle continuous infinity—they need finite steps. FDTD replaces derivatives with **central differences**. Instead of asking 'what is the exact slope at this point?', we ask 'what is the average slope between the point just ahead and the point just behind?' This approximation is remarkably accurate when our steps are small enough.

### 2. The Yee Grid: A Staggered Chessboard

The genius of Kane Yee's algorithm lies in how it arranges fields in space and time. Imagine a chessboard where:

- Electric fields (E) live on the black squares.
- Magnetic fields (H) live on the white squares.

In the Yee Grid, E and H are not computed at the same physical location or time. They are offset by half a grid cell in space and half a time step in time. This 'leapfrog' approach allows us to use simple, second-order accurate differences.

### 3. 1D Wave Propagation

For a simple wave traveling along the z-direction, we assume the electric field points in the y-direction and the magnetic field points in the x-direction. Maxwell's equations then simplify to two coupled equations:

$$\frac{\partial H_x}{\partial t} = \frac{1}{\mu} \cdot \frac{\partial E_y}{\partial z}$$

$$\frac{\partial E_y}{\partial t} = \frac{1}{\epsilon} \cdot \frac{\partial H_x}{\partial z}$$

Here,  $\epsilon$  (epsilon) is the material's permittivity and  $\mu$  (mu) is its permeability. In free space, these become  $\epsilon_0$  and  $\mu_0$ .

### 4. 2D TMz Polarization

In two dimensions, waves become more interesting. In the TMz mode, the electric field points only along the z-axis, and the magnetic field lies in the xy-plane. This gives us three coupled update equations instead of two, allowing us to model wave behavior in a plane.

## Solution Methodologies:

The methodologies used in this project:

- **FDTD (Finite-Difference Time-Domain):** The core numerical engine.
- **Yee Grid Staggering:** Spatial and temporal offsetting of E and H fields.
- **Central Difference Approximation:** Replacing derivatives with algebraic differences.
- **CFL Stability Criterion:** Ensuring the time step is small enough to prevent numerical explosion.
- **Mur Absorbing Boundary Conditions (ABC):** Preventing artificial reflections at grid edges.
- **Gaussian Pulse Source:** Injecting a wideband electromagnetic signal.
- **DFT (Discrete Fourier Transform):** Converting time-domain results to frequency-domain spectra.

## Algorithm:

### 1. Input Parameters:

Users define the simulation environment through the GUI or script: grid size ( $N_x$ ,  $N_y$ ), spatial steps ( $\Delta x$ ,  $\Delta y$ ), maximum frequency ( $f_{\max}$ ), and material properties ( $\epsilon$ ,  $\mu$ ) for each region.

### 2. Defining Simulation Parameters:

```
c0 = 3e8; % Speed of light in vacuum  
f_max = 1e9; % Maximum frequency (1 GHz)  
lambda_min = c0 / f_max;  
N_lambda = 50; % Cells per wavelength  
dx = lambda_min / N_lambda;  
dy = dx; % Uniform grid
```

### 3. Computing Grid Resolution:

The grid must be fine enough to resolve the shortest wavelength. Typically, we use at least 50 cells per wavelength to avoid wave distortion.

### 4. Computing Time Step (CFL Condition):

To keep the simulation stable, the time step must satisfy the Courant-Friedrichs-Lewy criterion. In our implementation, we set  $\Delta t$  exactly at this stability limit for maximum accuracy:

$$\Delta t \leq 1 / [c \cdot \sqrt{(1/\Delta x^2 + 1/\Delta y^2)}]$$

## 5. Computing Source:

A Gaussian pulse is used because it contains a wide range of frequencies simultaneously. The pulse is defined as:

$$E_{\text{src}}(t) = \exp[ -((t - t_{\text{delay}}) / (0.2\tau))^2 ]$$

where  $\tau = 0.5/f_{\text{max}}$  and  $t_{\text{delay}}$  is a delay ensuring the pulse starts smoothly from zero.

## 6. Computing Update Coefficients:

```
mEz = (c0 * dt) ./ eps_r; % Electric field coefficient  
mHx = (c0 * dt) ./ mu_r; % Magnetic field coefficient
```

## 7. Initializing Fourier Transform:

```
freq = 10.^((0:0.01:9)); % Frequency range  
Nfreq = length(freq);  
K = exp(-1j * 2 * pi * freq * dt);  
ref = zeros(1, Nfreq);  
trn = zeros(1, Nfreq);  
src = zeros(1, Nfreq);
```

## 8. Initializing Fields and Boundary Terms:

```
Ez = zeros(Nx, Ny); % Electric field (z-comp)  
Hx = zeros(Nx, Ny); % Magnetic field (x-comp)  
Hy = zeros(Nx, Ny); % Magnetic field (y-comp)
```

## 9. Main FDTD Loop:

i. Save Boundary Edges for Mur ABC:

```
Ez_left = Ez(2, :);  
  
Ez_right = Ez(Nx-1, :);  
  
Ez_bottom = Ez(:, 2);  
  
Ez_top = Ez(:, Ny-1);
```

ii. Update Magnetic Fields:

```
Hx(:, 1:Ny-1) = Hx(:, 1:Ny-1) -  
(mHx/dy)*(Ez(:, 2:Ny)-Ez(:, 1:Ny-1));  
  
Hy(1:Nx-1, :) = Hy(1:Nx-1, :) +  
(mHy/dx)*(Ez(2:Nx, :)-Ez(1:Nx-1, :));
```

iii. Inject Source:

Overwrite a specific grid node with the Gaussian pulse value at the current time step.

iv. Update Electric Field:

```
Ez(2:Nx-1, 2:Ny-1) = Ez(2:Nx-1, 2:Ny-1) + ...  
(mEz(2:Nx-1, 2:Ny-1)./dx).*(Hy(2:Nx-1, 2:Ny-1)-Hy(1:Nx-2, 2:Ny-1)  
) - ...  
(mEz(2:Nx-1, 2:Ny-1)./dy).*(Hx(2:Nx-1, 2:Ny-1)-Hx(2:Nx-1, 1:Ny-2)  
);
```

v. Apply Mur Absorbing Boundary Conditions:

```
Ez(1, :) = Ez_left;  
  
Ez(Nx, :) = Ez_right;  
  
Ez(:, 1) = Ez_bottom;  
  
Ez(:, Ny) = Ez_top;
```

vi. Update Fourier Transform:

```
for nf = 1:Nfreq  
    ref(nf) = ref(nf) + K(nf)^T * Ez(sensor_ref);  
    trn(nf) = trn(nf) + K(nf)^T * Ez(sensor_trn);  
    src(nf) = src(nf) + K(nf)^T * E_src(T);  
end
```

vii. Plotting Fields:

Real-time visualization of the 2D E-field wavefront propagating through the grid, along with current frequency-domain reflectance and transmittance curves.

viii. Computing Response:

```
Reflectance = abs(ref ./ src).^2;  
Transmittance = abs(trn ./ src).^2;  
Total = Reflectance + Transmittance;
```

ix. Exporting Results:

Upon completion, frequency, reflectance, transmittance, and conservation data are exported to a timestamped Excel file.

## Sample Test Case Results and Explanation:

### 1. Air – Glass – Air (1D Simulation)

A Gaussian pulse is injected in air and travels toward a glass slab. As the wave hits the air-glass boundary, part of it reflects back and part transmits through. Inside glass, the wave slows down (since glass has a higher refractive index) and its wavelength compresses. Upon exiting back into air, it speeds up again.

#### Observations:

- The time-domain plot shows the incident pulse, the reflected pulse traveling backward, and the transmitted pulse moving forward.
- The frequency-domain plot shows nearly constant reflectance (~4%) and transmittance (~96%) across the spectrum, matching the theoretical Fresnel coefficients for air-to-glass.

### 2. Air – Multi-Layer Device – Air (1D Simulation)

Here we simulate a three-layer device (e.g., Paper-Graphite-Glass). Each layer has different  $\epsilon$  and  $\mu$  values. The wave undergoes multiple internal reflections at each interface, creating complex interference patterns.

#### Observations:

- The time-domain plot shows ringing and multiple echoes within the device layer.
- The frequency-domain plot reveals spectral resonances—certain frequencies are strongly reflected while others pass through almost entirely. This is the principle behind optical filters and anti-reflection coatings.

### 3. 2D Wave Propagation in Free Space (2D TMz Simulation)

A point source radiates circular wavefronts in a 2D grid. The wave expands outward symmetrically like ripples on a pond.

#### Observations:

- The 2D contour plot shows perfect circular wavefronts in the early time steps.
- As the wave reaches the grid edges, the Mur ABC absorbs it cleanly with minimal reflection, simulating an open infinite space. Without ABC, the wave would bounce off the boundaries like a mirror, contaminating the entire simulation.

#### 4. 2D Wave Passing Through a Dielectric Cylinder

A plane wave is launched from the left side of the grid and encounters a circular dielectric object ( $\epsilon_r = 4$ ).

**Observations:**

- The wave bends (refracts) as it enters the cylinder.
- A shadow region forms behind the cylinder.
- Some energy diffracts around the edges, filling in the shadow partially. This demonstrates FDTD's ability to model complex scattering geometries that are impossible to solve analytically.

## Dealing with Problems:

### 1. Numerical Stability and the CFL Condition

If the time step  $\Delta t$  is too large compared to the spatial grid, the simulation becomes unstable and field values grow exponentially to infinity. The CFL criterion fixes this by ensuring that numerical information cannot travel faster than the speed of light across a single cell. We always set  $\Delta t$  at the stability limit to maximize accuracy while preventing blow-up.

### 2. Artificial Boundary Reflections

Computer memory is finite, so the grid must end somewhere. Without treatment, waves hitting the grid edge reflect back as if striking a perfect metal wall. The Mur ABC solves this by extrapolating the field value at the boundary from the adjacent inner cell's previous value, allowing the wave to exit gracefully.

### 3. Grid Resolution vs. Computation Time

Finer grids (more cells per wavelength) give more accurate results but require more memory and more time steps to simulate. We typically use 50 cells per wavelength as a practical compromise between accuracy and computational cost.

### 4. Dispersion Errors

In FDTD, different frequencies travel at slightly different numerical speeds. This means a broadband Gaussian pulse may spread out or distort over long distances. This error is minimized by using sufficiently fine grids.

### 5. Source Injection Location

The source must be injected carefully so that it does not create non-physical discontinuities. We use a soft source (adding the pulse to the existing field) rather than a hard source (overwriting the field completely) to allow scattered waves to pass through the source location.

### 6. E-H Amplitude Scaling

In free space, the intrinsic impedance  $\eta = \sqrt{\mu_0/\epsilon_0} \approx 377 \, \Omega$  means that the magnetic field amplitude is much smaller than the electric field. For visualization purposes, we often normalize H to match E's amplitude so both can be plotted on the same scale.

## Discussion and Future Work:

This project provides a comprehensive foundation for understanding the FDTD method. We have successfully deconstructed Maxwell's continuous equations into discrete, computable steps and implemented a stable solver with proper boundary conditions and spectral analysis.

However, there is significant room for advancement:

**Three-Dimensional Extension:** Real-world electromagnetic problems are inherently 3D. Extending the 2D TMz formulation to full 3D would allow modeling of antennas, complex circuits, and biological tissue interactions.

**Advanced Material Modeling:** Current simulations use simple, non-dispersive materials. Future work could include dispersive materials (where  $\epsilon$  varies with frequency), anisotropic materials (where properties depend on direction), and nonlinear materials.

**Perfectly Matched Layer (PML):** While Mur ABC works well for normal incidence, the Berenger PML absorbs waves at all angles with near-perfect efficiency. Replacing Mur ABC with PML would dramatically improve simulation accuracy for oblique incidence.

**GPU Acceleration:** FDTD is embarrassingly parallel—each grid cell updates independently. Porting the MATLAB code to CUDA or OpenCL could speed up large simulations by orders of magnitude.

**Multi-Physics Coupling:** Electromagnetic fields often interact with heat (microwave heating) or mechanical stress. Coupling FDTD with thermal solvers would open doors to microwave engineering and medical applications.

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